

Christopher Johnstone

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Education

Sep 2019 – Ongoing	Massachusetts Institute of Technology, Cambridge, MA <i>PhD candidate in Chemical Engineering, minor in biological systems modeling.</i> • Advisor: Kate E. Galloway • Thesis: Supercoiling as a regulator: the role of biophysical feedback in transcription and chromatin dynamics.
Aug 2015 – Jun 2019	California Institute of Technology, Pasadena, CA <i>BS in Chemical Engineering, Minor in Computer Science.</i> • Advisor: David A. Tirrell • Senior thesis: Control of aggregated bacterial communities through engineered surface displayed proteins.

Experience

Dec 2019 – Ongoing	Galloway Lab, MIT – Cambridge, MA <i>PhD candidate</i> • Investigating the impact of transcription-induced supercoiling on gene regulation.
Fall 2021	Dept. of Chemical Engineering, MIT – Cambridge, MA <i>TA for Graduate Thermodynamics</i> • Developed homeworks, wrote course exams, prepared review sessions, and wrote lecture notes.
Dec 2016 – Jun 2019	Tirrell Lab, Caltech – Pasadena, MA <i>Research student, Robb and Eunice Rutledge SURF fellow (summer 2017)</i> • Demonstrated a synthetic aggregation circuit and quorum sensing circuit to manipulate and sense aggregation of communities of <i>E. coli</i> cells. • Rationally designed and characterized a protein for photocontrollable bacterial aggregation.
2017-2019	Dept. of Chemical Engineering, Dept. of Computer Science, Caltech – Pasadena, CA <i>TA for Separation Processes, Intro to Chemical Engineering Computation, Computer Graphics, Intro to Programming Methods</i> • Developed novel examples, practice tests, and review sessions for weekly recitation sessions.
Jun 2018 – Aug 2018	Provivi – Santa Monica, CA <i>Metabolic engineering intern</i> • Identified and overexpressed bottleneck pathway genes, creating a yeast strain producing 30% more lipid dry mass.
Jun 2016 – Sep 2016	Pure Storage – Mountain View, CA <i>Software engineering intern</i> • Implemented a customer-facing feature to accurately account utilized storage space across clusters.

Publications

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| 2025 | 1. Johnstone, C. P. et al. Gene Syntax Defines Supercoiling-Mediated Transcriptional Feedback. <i>bioRxiv</i> (Jan. 19, 2025). Pre-published. |
| | 2. Love, K. S., Johnstone, C. P. , ... & Galloway, K. E. Model-Guided Design of microRNA-based Gene Circuits Supports Precise Dosage of Transgenic Cargoes into Diverse Primary Cells. <i>Cell Systems</i> (June 18, 2025). |
| | 3. Peterman, E. L., ..., Johnstone, C. P. , & Galloway, K. E. High-Resolution Profiling Reveals Coupled Transcriptional and Translational Regulation of Transgenes. <i>Nucleic Acids Research</i> (June 6, 2025). |
| 2023 | 4. Kozlowski, M. T., ..., Johnstone, C. P. , ..., Tirrell, D. A. & Ku, H. T. A Matrigel-Free Method for Culture of Pancreatic Endocrine-like Cells in Defined Protein-Based Hydrogels. <i>Frontiers in Bioengineering and Biotechnology</i> (Mar. 9, 2023). |
| 2022 | 5. Johnstone, C. P. & Galloway, K. E. Supercoiling-Mediated Feedback Rapidly Couples and Tunes Transcription. <i>Cell Reports</i> (Oct. 18, 2022). |
| 2021 | 6. Johnstone, C. P. & Galloway, K. E. Engineering Cellular Symphonies out of Transcriptional Noise. <i>Nature Reviews Molecular Cell Biology</i> (Mar. 15, 2021). |
| | 7. Kozlowski, M. T., Silverman, B. R., Johnstone, C. P. & Tirrell, D. A. Genetically Programmable Microbial Assembly. <i>ACS Synthetic Biology</i> (June 18, 2021). |
| 2020 | 8. Johnstone, C. P. , Wang, N. B., Sevier, S. A. & Galloway, K. E. Understanding and Engineering Chromatin as a Dynamical System across Length and Timescales. <i>Cell Systems</i> (Nov. 18, 2020). |

Talks

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| 2025 | 1. Genome Architecture and Function (Poster). <i>Gene syntax defines supercoiling-mediated transcriptional feedback</i> . Sofia, Bulgaria. |
| | 2. Engineering Biology Club, Ginkgo Bioworks. <i>Gene syntax defines supercoiling-mediated transcriptional feedback</i> . Boston, MA. |
| 2024 | 1. March APS. <i>Supercoiling-mediated feedback rapidly couples and tunes transcription in mammalian cells</i> . Minneapolis, MN. |
| 2023 | 2. Boston Mammalian Synthetic Biology. <i>Supercoiling reversibly couples adjacent gene expression through biophysical feedback</i> . Boston, MA |
| | 3. Mammalian Synthetic Biology Workshop (Poster). <i>Supercoiling-mediated transcriptional dynamics impart rapid, tunable feedback</i> . San Jose, CA. |
| 2022 | 4. AIChE Annual Meeting. <i>Modeling Supercoiling-Dependent Feedback As a Transcriptional Coordinator to Understand and Engineer Biological Circuits</i> . Phoenix, AZ. |
| | 5. Boston Mammalian Synthetic Biology. <i>Supercoiling-mediated feedback rapidly couples and tunes transcription</i> . Boston, MA. |
| 2018 | 6. 255th ACS National Meeting. Poster. <i>Strategies for controlled bacterial assembly resulting in activation of a quorum-sensing circuit</i> . (with M. T. Kozlowski). New Orleans, LA. |

Leadership

Mar 2020 – Ongoing	Sidney Pacific Graduate Residence, MIT <i>VP of Residential Life, Web Chair Lead</i> Secured funding for the residence, and led a group of officers in development of safe social events to build community amid pandemic restrictions. Led the team that develops software for residents and building staff.
Sep 2020 – Sep 2021	GSCX, MIT Coordinated and organized virtual social events for the Chemical Engineering department throughout Covid.
Apr 2016 – Apr 2019	Board of Control, Caltech <i>Board member, Secretary (until Mar 2018), Chair</i> Led investigations, coordinated with administrators and professors, and organized hearings for academic honor code violations.

Honors

Jun 2019	Frederic W. Hinrichs Jr. Memorial Award Recipient Award given to the two seniors who have made the greatest contribution to the student body and whose qualities of character, leadership, and responsibility have been outstanding.
May 2018	Donald S. Clark Award Recipient Award given to two engineering juniors who show academic excellence, leadership, and dedication to the Caltech community.
Mar 2018	Elected to Tau Beta Pi

Skills

- Experimental:** Tissue culture, iPSC culture, lentivirus production, molecular cloning, genomics (sci-ATAC-seq, Micro-C, CUT&Tag, NGS library prep), FACS and flow cytometry, protein production and purification. Basic mouse training.
- Computational:** Biological circuit modeling, stochastic simulations, bioinformatic and NGS pipelines. Computer graphics and networking programming experience. Extensive experience in Python, Julia, C++, C. Experience in Go, Haskell, MATLAB, TypeScript & JS, PHP.